

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Chapter 6: The Riemann–Stieltjes Integral

Section: Properties of the Integral

Based on Walter Rudin's *Principles of Mathematical Analysis*

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Outline

$$\int_a^b f(x) dg(x)$$

Motivation

Algebraic Properties

Products and Absolute Values

Step Functions: Series as Integrals

Reducing to the Riemann Integral

Why Stieltjes?

Change of Variable

Summary



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Motivation



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

From Existence to Rules

- $\int_a^b f(x) dg(x)$
- Last episode: **when** does $\int_a^b f d\alpha$ exist? (Theorem 6.6: $U - L < \varepsilon$)



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

From Existence to Rules

$$\int_a^b f(x) dg(x)$$

- Last episode: **when** does $\int_a^b f d\alpha$ exist? (Theorem 6.6: $U - L < \varepsilon$)
- This episode: **how does the integral behave?** Linearity, order, splitting, products, absolute values



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

From Existence to Rules

- Last episode: **when** does $\int_a^b f d\alpha$ exist? (Theorem 6.6: $U - L < \varepsilon$)
- This episode: **how does the integral behave?** Linearity, order, splitting, products, absolute values
- The payoff of the Stieltjes viewpoint:

$$\text{step weight } \alpha \Rightarrow \int f d\alpha = \sum_n c_n f(s_n) \quad (\text{series!})$$

$$\alpha' \in \mathcal{R} \Rightarrow \int f d\alpha = \int f \alpha' dx \quad (\text{Riemann!})$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Our Workhorse from Last Time

Recall: Theorem 6.6 (the integrability criterion)

$f \in \mathcal{R}(\alpha)$ on $[a, b] \iff$ for every $\varepsilon > 0$ there is a partition P with

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon. \quad (13)$$

Recall: Theorem 6.7 (its companions)

- (a) If (13) holds for P , it holds for every **refinement** of P .
- (b) $\sum_i |f(s_i) - f(t_i)| \Delta\alpha_i < \varepsilon$ for sample points $s_i, t_i \in [x_{i-1}, x_i]$.
- (c) If $f \in \mathcal{R}(\alpha)$: $|\sum_i f(t_i) \Delta\alpha_i - \int_a^b f d\alpha| < \varepsilon$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Algebraic Properties



Theorem 6.12(a): Linearity in the Function

Theorem 6.12(a)

If $f_1 \in \mathcal{R}(\alpha)$ and $f_2 \in \mathcal{R}(\alpha)$ on $[a, b]$, then $f_1 + f_2 \in \mathcal{R}(\alpha)$, $cf \in \mathcal{R}(\alpha)$ for every constant c , and

$$\int_a^b (f_1 + f_2) d\alpha = \int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha, \quad \int_a^b cf d\alpha = c \int_a^b f d\alpha.$$

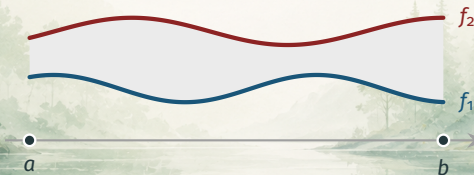
Key idea: integration is a **linear** operation on $\mathcal{R}(\alpha)$.

Theorem 6.12(b): Monotonicity

Theorem 6.12(b)

If $f_1(x) \leq f_2(x)$ on $[a, b]$, then

$$\int_a^b f_1 d\alpha \leq \int_a^b f_2 d\alpha.$$



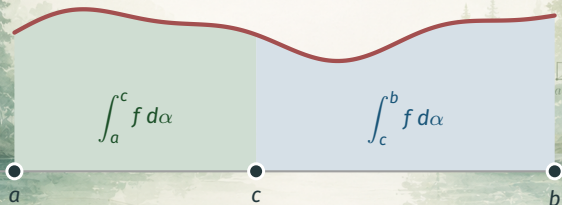
Key idea: the integral respects order.

Theorem 6.12(c): Splitting the Interval

Theorem 6.12(c)

If $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and $a < c < b$, then $f \in \mathcal{R}(\alpha)$ on $[a, c]$ and on $[c, b]$, and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

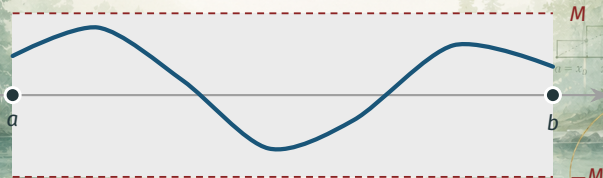


Theorem 6.12(d): The Basic Size Bound

Theorem 6.12(d)

If $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and $|f(x)| \leq M$ on $[a, b]$, then

$$\left| \int_a^b f d\alpha \right| \leq M [\alpha(b) - \alpha(a)].$$



Theorem 6.12(e): Linearity in the Weight

Theorem 6.12(e)

If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and

$$\int_a^b f d(\alpha_1 + \alpha_2) = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2;$$

if $f \in \mathcal{R}(\alpha)$ and $c > 0$ is constant, then $f \in \mathcal{R}(c\alpha)$ and

$$\int_a^b f d(c\alpha) = c \int_a^b f d\alpha.$$

Key idea: the integral is linear in the function f and the weight α .

Proof of Theorem 6.12(a) — Step 1: The Key Inequality

Step 1: Sums of functions vs. sums of sums

Let $f = f_1 + f_2$. For any partition P of $[a, b]$:

$$L(P, f_1, \alpha) + L(P, f_2, \alpha) \leq L(P, f, \alpha) \leq U(P, f, \alpha) \leq U(P, f_1, \alpha) + U(P, f_2, \alpha). \quad (20)$$

Why: on each subinterval, $\sup(f_1 + f_2) \leq \sup f_1 + \sup f_2$ and $\inf(f_1 + f_2) \geq \inf f_1 + \inf f_2$.

Proof of Theorem 6.12(a) — Step 2: $f_1 + f_2$ Is Integrable

Step 2: Apply the criterion

Given $\varepsilon > 0$, choose partitions P_j ($j = 1, 2$) with

$$U(P_j, f_j, \alpha) - L(P_j, f_j, \alpha) < \varepsilon.$$

These persist for the **common refinement** P of P_1, P_2 . Then (20) gives

$$U(P, f, \alpha) - L(P, f, \alpha) < 2\varepsilon \quad \implies \quad f \in \mathcal{R}(\alpha).$$

Technique: one partition must work for **both** functions — take the common refinement.

Proof of Theorem 6.12(a) — Step 3: The Value from Above

Step 3: Compare with the sum of the integrals

With this same P ,

$$U(P, f_j, \alpha) < \int f_j d\alpha + \varepsilon \quad (j = 1, 2);$$

hence (20) implies

$$\int f d\alpha \leq U(P, f, \alpha) < \int f_1 d\alpha + \int f_2 d\alpha + 2\varepsilon.$$

Since ε was arbitrary,

$$\int f d\alpha \leq \int f_1 d\alpha + \int f_2 d\alpha. \quad (21)$$

Proof of Theorem 6.12(a) — Step 4: The Value from Below

Step 4: Run Step 3 with *lower* sums

With the same P : $L(P, f_j, \alpha) > \int f_j d\alpha - \varepsilon$ ($j = 1, 2$). The **left** half of (20) then gives

$$\int f d\alpha \geq L(P, f, \alpha) \geq L(P, f_1, \alpha) + L(P, f_2, \alpha) > \int f_1 d\alpha + \int f_2 d\alpha - 2\varepsilon.$$

Since ε was arbitrary, this **reverses** (21):

$$\int (f_1 + f_2) d\alpha = \int f_1 d\alpha + \int f_2 d\alpha$$

□

Technique: (20) is a **two-sided** sandwich — use each half once.

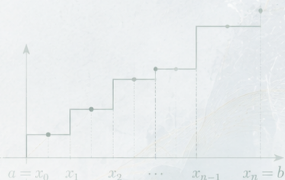
Theorem 6.12: The Remaining Parts

Rudin: “The proofs are so similar that we omit the details”

- the cf half of (a), and (b), (d), (e): run the **same pattern** — compare sums, apply Theorem 6.6, squeeze with ε
- (c): the point is to restrict to partitions that **contain the point c** (pass to refinements!); each such P splits into a partition of $[a, c]$ and one of $[c, b]$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Products and Absolute Values

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

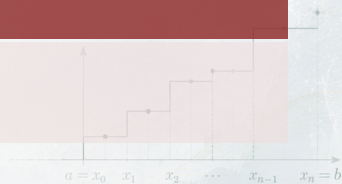
Theorem 6.13: Products and Absolute Values

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.13

If $f \in \mathcal{R}(\alpha)$ and $g \in \mathcal{R}(\alpha)$ on $[a, b]$, then

(a) $fg \in \mathcal{R}(\alpha)$;



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.13: Products and Absolute Values

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.13

If $f \in \mathcal{R}(\alpha)$ and $g \in \mathcal{R}(\alpha)$ on $[a, b]$, then

(a) $fg \in \mathcal{R}(\alpha)$;

(b) $|f| \in \mathcal{R}(\alpha)$ and $\left| \int_a^b f d\alpha \right| \leq \int_a^b |f| d\alpha$.



Key idea: (b) is the **triangle inequality for integrals** — an integral is a limit of sums, and sums obey $|x + y| \leq |x| + |y|$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.13(a) — Step 1: Squares

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Recall: Theorem 6.11 (composition)

If $f \in \mathcal{R}(\alpha)$, $m \leq f \leq M$, and ϕ is continuous on $[m, M]$, then $h = \phi \circ f \in \mathcal{R}(\alpha)$.

Step 1: $f \in \mathcal{R}(\alpha) \Rightarrow f^2 \in \mathcal{R}(\alpha)$

Take $\phi(t) = t^2$ (continuous!). Theorem 6.11 gives

$$f \in \mathcal{R}(\alpha) \implies f^2 = \phi \circ f \in \mathcal{R}(\alpha).$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.13(a) — Step 2: The Polarization Trick

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Step 2: Products from squares

$$4fg = (f + g)^2 - (f - g)^2.$$

- $f + g, f - g \in \mathcal{R}(\alpha)$
- $(f + g)^2, (f - g)^2 \in \mathcal{R}(\alpha)$
- $4fg \in \mathcal{R}(\alpha)$, hence $fg \in \mathcal{R}(\alpha)$

[Theorem 6.12(a)]

[Step 1]

[Theorem 6.12(a)]



Technique: the **polarization identity**:

recover a product from squares of sums and differences.

Proof of Theorem 6.13(b) — Step 1: $|f|$ Is Integrable

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Step 1: Compose with $\phi(t) = |t|$

The absolute value function is continuous, so Theorem 6.11 gives

$$f \in \mathcal{R}(\alpha) \implies |f| = \phi \circ f \in \mathcal{R}(\alpha).$$

Same trick, different ϕ : yesterday t^2 , today $|t|$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.13(b) — Step 2: One Well-Chosen Sign

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Step 2: Choose $c = \pm 1$

Pick $c = \pm 1$ so that $c \int f d\alpha \geq 0$. Then

$$\left| \int f d\alpha \right| = c \int f d\alpha = \int cf d\alpha \leq \int |f| d\alpha,$$

since $cf \leq |f|$. [6.12(a) for the middle step, 6.12(b) for the last] $\square$

Technique: a single sign choice replaces a two-case argument.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Step Functions: Series as Integrals



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

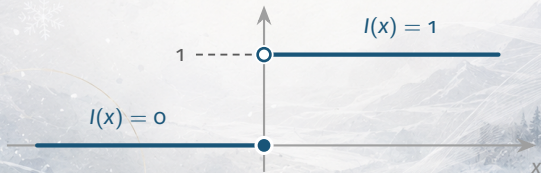
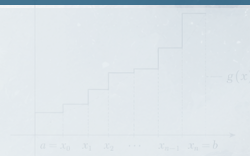
Definition 6.14: The Unit Step Function

Definition 6.14

The **unit step function** I is defined by

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$I(x) = \begin{cases} 0 & (x \leq 0), \\ 1 & (x > 0). \end{cases}$$



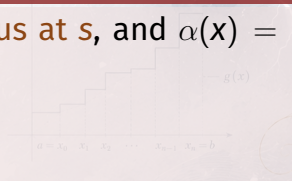
$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.15: Integration Against a Single Jump

Theorem 6.15

If $a < s < b$, f is bounded on $[a, b]$, f is **continuous at s** , and $\alpha(x) = I(x - s)$, then

$$\int_a^b f d\alpha = f(s).$$



Key idea: all the weight sits at s — the integral **samples f** there. (Physicists: this is the “ δ -function” made rigorous.)

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.15 — Only One Interval Counts

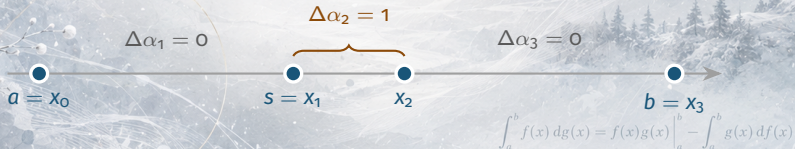
A four-point partition

Consider $P = \{x_0, x_1, x_2, x_3\}$ with $x_0 = a$, $x_1 = s < x_2 < x_3 = b$. The increments of $\alpha(x) = I(x - s)$:

$$\Delta\alpha_1 = 0, \quad \Delta\alpha_2 = 1, \quad \Delta\alpha_3 = 0.$$

Hence

$$U(P, f, \alpha) = M_2, \quad L(P, f, \alpha) = m_2.$$



Proof of Theorem 6.15 — Shrink the Interval

Let $x_2 \rightarrow s$

f is continuous at s , so

$$M_2 = \sup_{[s, x_2]} f \longrightarrow f(s) \quad \text{and} \quad m_2 = \inf_{[s, x_2]} f \longrightarrow f(s) \quad (x_2 \rightarrow s).$$

Since $L(P, f, \alpha) \leq \underline{\int} f d\alpha \leq \overline{\int} f d\alpha \leq U(P, f, \alpha)$ for these P :

$$\int_a^b f d\alpha = f(s).$$

□

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.16: A Series Is an Integral

Theorem 6.16

Suppose $c_n \geq 0$, $\sum c_n$ converges, $\{s_n\}$ is a sequence of **distinct** points in (a, b) , and

$$\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n). \quad (22)$$

Let f be continuous on $[a, b]$. Then

$$\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n). \quad (23)$$

Key idea: a **series** is a Stieltjes integral against a pure step function.

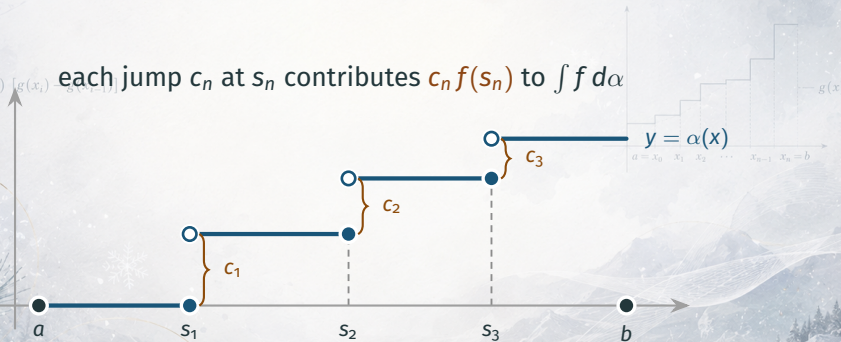
$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

The Picture: A Staircase Weight

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

each jump c_n at s_n contributes $c_n f(s_n)$ to $\int f d\alpha$



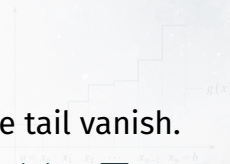
$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof Strategy: Theorem 6.16

$$\int_a^b f(x) dg(x)$$

Goal: $\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Approach: split the weight, use what we know, let the tail vanish.

1. Check α is well defined, increasing, with $\alpha(b) - \alpha(a) = \sum c_n$.
2. Split $\alpha = \alpha_1 + \alpha_2$: the **first N jumps** plus a **small tail**.
3. Finite part: Theorems 6.12(e) + 6.15 $\Rightarrow$ a finite sum.
4. Tail: total weight $< \varepsilon$, so Theorem 6.12(d) kills it.
5. Let $N \rightarrow \infty$.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.16 — Step 1: Split the Weight

Step 1: Setup and splitting

Comparison test: (22) converges for every x ;
 α is monotonic with $\alpha(a) = 0$, $\alpha(b) = \sum c_n$.

Given $\varepsilon > 0$, choose N with $\sum_{n=N+1}^{\infty} c_n < \varepsilon$, and put

$$\alpha_1(x) = \sum_{n=1}^N c_n I(x - s_n), \quad \alpha_2(x) = \sum_{n=N+1}^{\infty} c_n I(x - s_n).$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Proof of Theorem 6.16 — Step 2: Finite Part and Tail

Step 2: Integrate each piece

By Theorems 6.12(e) and 6.15 (applied N times),

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$\int_a^b f d\alpha_1 = \sum_{n=1}^N c_n f(s_n). \quad (24)$$

Since $\alpha_2(b) - \alpha_2(a) < \varepsilon$, Theorem 6.12(d) gives

$$\left| \int_a^b f d\alpha_2 \right| \leq M\varepsilon, \quad M = \sup |f(x)|. \quad (25)$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

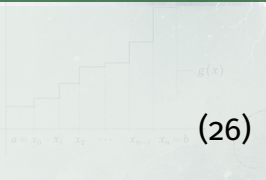
Proof of Theorem 6.16 — Step 3: Let $N \rightarrow \infty$

Step 3: Assemble and pass to the limit

Since $\alpha = \alpha_1 + \alpha_2$, (24) and (25) give

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$\left| \int_a^b f d\alpha - \sum_{n=1}^N c_n f(s_n) \right| \leq M\varepsilon. \quad (26)$$



Let $N \rightarrow \infty$: the partial sums converge to the series, and (23) follows. $\square$

$$\boxed{\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)} \quad \text{— series are integrals.}$$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

Reducing to the Riemann Integral



$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Theorem 6.17: When the Weight Has a Derivative

Theorem 6.17

Assume α increases monotonically and $\alpha' \in \mathcal{R}$ on $[a, b]$. Let f be a bounded real function on $[a, b]$. Then $f \in \mathcal{R}(\alpha)$ **if and only if** $f\alpha' \in \mathcal{R}$. In that case

$$\int_a^b f d\alpha = \int_a^b f(x) \alpha'(x) dx. \quad (27)$$

Key idea: $d\alpha = \alpha'(x) dx$ — the familiar substitution rule.

Proof Strategy: Theorem 6.17

$$f(x) dg(x)$$

Goal: show the upper integrals of $f d\alpha$ and of $f\alpha' dx$ coincide (then do the same for lower).

Approach:

1. Apply Theorem 6.6 to α' : get P with $U(P, \alpha') - L(P, \alpha') < \varepsilon$.
2. **Mean value theorem:** $\Delta\alpha_i = \alpha'(t_i) \Delta x_i$ for some t_i in each subinterval.
3. Theorem 6.7(b): swapping t_i for any s_i costs less than ε .
4. Conclude the two kinds of Riemann sums differ by $\leq M\varepsilon$; pass to upper sums, then refinements, then upper integrals.



Proof of Theorem 6.17 — Step 1: Control α'

Step 1: A good partition for α' , then MVT

Given $\varepsilon > 0$, Theorem 6.6 (applied to α') gives a partition $P = \{x_0, \dots, x_n\}$ with

$$U(P, \alpha') - L(P, \alpha') < \varepsilon. \quad (28)$$

The **mean value theorem** furnishes $t_i \in [x_{i-1}, x_i]$ with

$$\Delta\alpha_i = \alpha'(t_i) \Delta x_i \quad (i = 1, \dots, n).$$

If $s_i \in [x_{i-1}, x_i]$, then by (28) and Theorem 6.7(b),

$$\sum_{i=1}^n |\alpha'(s_i) - \alpha'(t_i)| \Delta x_i < \varepsilon. \quad (29)$$


$$\int_a^b f(x) dg(x)$$

$$\sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Proof of Theorem 6.17 — Step 2: Compare the Two Sums

Step 2: Stieltjes sums vs. Riemann sums

Put $M = \sup |f(x)|$. The MVT relation gives

$$\sum_{i=1}^n f(s_i) \Delta \alpha_i = \sum_{i=1}^n f(s_i) \alpha'(t_i) \Delta x_i,$$

so (29) implies

$$\left| \sum_{i=1}^n f(s_i) \Delta \alpha_i - \sum_{i=1}^n f(s_i) \alpha'(s_i) \Delta x_i \right| \leq M \varepsilon. \quad (30)$$

Same s_i in both sums — only α' 's argument moved.

$$\int_a^b f(x) dg(x)$$



Proof of Theorem 6.17 — Step 3: Upper Sums Are Close

Step 3: From sums to upper sums

By (30), for all choices of $s_i \in [x_{i-1}, x_i]$:

$$\sum_{i=1}^n f(s_i) \Delta\alpha_i \leq U(P, f\alpha') + M\varepsilon \quad \implies \quad U(P, f, \alpha) \leq U(P, f\alpha') + M\varepsilon.$$

The same argument gives $U(P, f\alpha') \leq U(P, f, \alpha) + M\varepsilon$. Thus

$$\left| U(P, f, \alpha) - U(P, f\alpha') \right| \leq M\varepsilon. \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})] \quad (31)$$

Proof of Theorem 6.17 — Step 4: Pass to the Integrals

Step 4: Refine, then squeeze

(28) survives refinement (Theorem 6.7(a)), hence so does (31). Taking infima over all refinements:

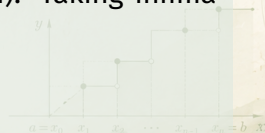
$$\left| \overline{\int_a^b f d\alpha} - \overline{\int_a^b f(x) \alpha'(x) dx} \right| \leq M\varepsilon.$$

But ε is arbitrary, so

$$\overline{\int_a^b f d\alpha} = \overline{\int_a^b f(x) \alpha'(x) dx} \quad \text{for any bounded } f. \quad (32)$$

Lower integrals: identical, from (30). The theorem follows. $\square$

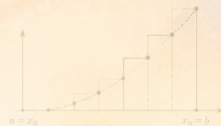
$$\int_a^b f(x) dg(x)$$



$$\sum_{i=0}^{n-1} f(\xi_i) [g(x_i) - g(x_{i-1})]$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

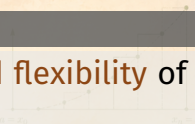


Why Stieltjes?

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Remark 6.18: One Theory for Series and Integrals

$i=1$



Remark 6.18

The two preceding theorems show the **generality and flexibility** of the Stieltjes process:

- α a **pure step function** $\Rightarrow$ the integral is a finite or infinite **series**
 - $\alpha' \in \mathcal{R} \Rightarrow$ the integral is an ordinary **Riemann integral**
- $\Rightarrow$ series and integrals can be studied **simultaneously**, not separately.

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

A Physical Example: Moment of Inertia

$t=1$

A wire on an axis

A straight wire of unit length spins about an axis through one endpoint, at right angles to the wire. Let $m(x)$ = mass contained in $[0, x]$. The **moment of inertia** is

$$\int_0^1 x^2 dm. \quad (33)$$

$$\int_a^b f(x) dg(x) = f(x)g(x)\Big|_a^b - \int_a^b g(x) df(x)$$

A Physical Example: Moment of Inertia

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$$\int_0^1 x^2 dm. \quad (33)$$

Continuous density $m' = \rho$:

$$\int_0^1 x^2 \rho(x) dx \quad (34)$$



Point masses m_i at x_i :

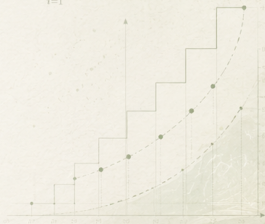
$$\sum_i x_i^2 m_i \quad (35)$$



$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$



Change of Variable

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Theorem 6.19: Change of Variable

Theorem 6.19

Suppose φ is a **strictly increasing continuous** function mapping $[A, B]$ onto $[a, b]$. Suppose α increases monotonically on $[a, b]$ and $f \in \mathcal{R}(\alpha)$ on $[a, b]$. Define β and g on $[A, B]$ by

$$\beta(y) = \alpha(\varphi(y)), \quad g(y) = f(\varphi(y)). \quad (36)$$

Then $g \in \mathcal{R}(\beta)$ and

$$\int_A^B g \, d\beta = \int_a^b f \, d\alpha. \quad (37)$$

$$\int_a^b f(x) \, dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) \, df(x)$$

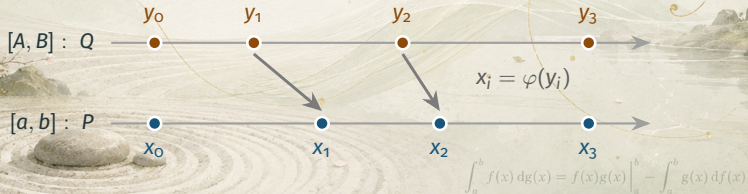
Proof of Theorem 6.19 — Partitions Correspond

Proof

Partitions $P = \{x_0, \dots, x_n\}$ of $[a, b]$ correspond to partitions $Q = \{y_0, \dots, y_n\}$ of $[A, B]$ via $x_i = \varphi(y_i)$ — **all** of them arise this way. f on $[x_{i-1}, x_i]$ takes the **same values** as g on $[y_{i-1}, y_i]$, so

$$U(Q, g, \beta) = U(P, f, \alpha), \quad L(Q, g, \beta) = L(P, f, \alpha). \quad (38)$$

Choose P making both close to $\int f d\alpha$ (Theorem 6.6);
Then (38) gives $g \in \mathcal{R}(\beta)$ and (37). □



The Classical Substitution Rule

Special case of (37)

Take $\alpha(x) = x$. Then $\beta = \varphi$. Assume $\varphi' \in \mathcal{R}$ on $[A, B]$. Applying Theorem 6.17 to the left side of (37):

$$\int_a^b f(x) dx = \int_A^B f(\varphi(y)) \varphi'(y) dy. \quad (39)$$

The calculus workhorse “ u -substitution” = Theorem 6.19 (relabel) + Theorem 6.17 ($d\varphi = \varphi' dy$).

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

$$\int_a^b f(x) dg(x)$$

$$S(P) = \sum_{i=1}^n f(\xi_i) [g(x_i) - g(x_{i-1})]$$

Summary

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Summary

$$\int_a^b f(x) dg(x)$$

$$S(x) = \sum_{i=1}^n f(s_i) [g(x_i) - g(x_{i-1})]$$

Key takeaways from this section:

- **Theorem 6.12:** the integral is **linear** in f and in α , respects order, splits at any $c \in (a, b)$, and obeys $|\int f d\alpha| \leq M[\alpha(b) - \alpha(a)]$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Summary

$$\int_a^b f(x) dg(x)$$

$$S(x) = \sum_{i=1}^n f(s_i) [g(x_i) - g(x_{i-1})]$$

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- **Theorem 6.13:** fg and $|f|$ stay integrable (polarization + composition), and $|\int f d\alpha| \leq \int |f| d\alpha$

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Summary

$$\int_a^b f(x) dg(x)$$

$$S(x) = \sum_{i=1}^n f(s_i) [g(x_i) - g(x_{i-1})]$$

Key takeaways from this section:

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- **Theorem 6.13:** fg and $|f|$ stay integrable (polarization + composition), and $|\int f d\alpha| \leq \int |f| d\alpha$
- **Theorems 6.14–6.16:** step-function weights **sample** f : $\int f d\alpha = \sum c_n f(s_n)$: a series **is** an integral

$$\int_a^b f(x) dg(x) = f(x)g(x) \Big|_a^b - \int_a^b g(x) df(x)$$

Summary

$$\int_a^b f(x) dg(x)$$

$$g(x) = \sum_{i=1}^n f(s_i) [g(x_i) - g(x_{i-1})]$$

Key takeaways from this section:

- **Theorem 6.12:** the integral is **linear** in f and in α , respects order, splits at any $c \in (a, b)$, and obeys $|\int f d\alpha| \leq M[\alpha(b) - \alpha(a)]$
- **Theorem 6.13:** fg and $|f|$ stay integrable (polarization + composition), and $|\int f d\alpha| \leq \int |f| d\alpha$
- **Theorems 6.14–6.16:** step-function weights **sample** f : $\int f d\alpha = \sum c_n f(s_n)$: a series **is** an integral
- **Theorem 6.17:** $\alpha' \in \mathcal{R} \Rightarrow d\alpha = \alpha' dx$ — back to Riemann; **6.19:** change of variable, no derivative needed

Coming up next: The **Fundamental Theorem of Calculus.**

$$\int_a^b f(x) dg(x) = \left. f(x)g(x) \right|_a^b - \int_a^b g(x) df(x)$$